

SALTDRAW · CALCULATION

SaltDrag sizing calculations

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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Source	docs/04-calcs/01-sizing.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-03	Amish Chadha	Rerun on the constructable design (SDG-DDR-002); pawl boss, ballast and lift-off near the bund added
0.1	2026-10-03	Amish Chadha	First sizing note at TRL 3 (drag, capstan, strength, floor, rate, mass and cost)

SaltDrag sizing calculations

Two people at 89 N each on 330 mm cranks can pull a 3 m rake with a full 230 kg heap across a 30 m pan in under ten minutes, and every load-bearing part keeps at least a 2.1 margin on yield at twice the rated 3 kN pull. Eleven of twelve requirements are met on paper. R5, the gathering rate against the dantala, cannot be shown until the dantala baseline is measured; the estimate is 150 kg per worker-hour. The largest uncertainty is the drag of a rake on a real crystal bed, which is assumed here and must be measured before the first build is loaded.

■ Scope and method

First-order hand calculations, scripted in `docs/04-calcs/sizing.py`, which reads its geometry and masses from the parametric model (`cad/src/model.py`) and the bill of materials, prints every result with a label in square brackets, and writes `docs/04-calcs/results.csv`. Strength checks use twice the rated rope pull (6 kN), as R3 asks. All results are estimates for TRL 3; none is a test result.

■ Assumptions

Table 1. Assumptions (every one is in the script's ASSUME list).

Assumption	Value	Basis
Loose salt bulk density	1,200 kg/m ³	Typical 1,100 to 1,300 for crystallised salt
Heap repose angle; heap sliding friction	35 degrees; 0.70	Granular salt; friction taken as tan 35 degrees
Allowance to loosen and lift the harvest layer	300 N per metre of blade	Estimate; to be measured with the partner (PRB open questions)
Skids on wet salt	friction 0.30	UHMW-PE on a wet crystalline surface
Back tension from the far capstan	200 N	Enough to keep the rake square
Efficiencies	roller 0.98, one spur stage 0.96, bearings 0.98	Greased gears with some dust
People	two crankers at up to 120 N; 30 turns a minute under load, 50 on the return	R2; sustained cranking
Moving to the next strip	8 min	Unchain, slide both capstans 3 m, move the rollers, re-anchor
Crew	four	Two cranking near, one at the far capstan, one with rollers and ropes
Rope	12 mm polyester double braid, 27 kN minimum breaking strength, splices 85 %	Catalogue class; certificate to confirm
Steels	C45 yield 340 MPa; S275 yield 275 MPa	Normalised bar; structural plate
FRP beam	flexural strength 200 MPa, modulus 23 GPa	Lower-bound pultruded values
Soil behind the bund	firm clay, undrained shear strength 50 kPa; helix bearing factor 9	To confirm by a pull test on site
Brake lining	friction 0.30	Woven lining on steel, dusty
Design pan	30 m between bund crests	Assumption, to be surveyed

■ A. Drag on the rake (R2, R3)

The rake is sized so that its design drag with a full heap matches the rated 3 kN pull.

- Full heap: a 300 mm blade holds a heap of triangular section, 0.0643 m² over 3.0 m, or 0.19 m³ [A1], about 231 kg [A2].
- Sliding the full heap over the bed: $0.70 \times 231 \text{ kg} \times 9.81 = 1,589 \text{ N}$ [A3].
- Loosening and lifting the harvest layer: $300 \text{ N/m} \times 3.0 \text{ m} = 900 \text{ N}$ [A4].
- Skids: the rake weighs 40.4 kg [A5] and carries two ballast bags of 23.8 kg each [A5b]; $0.30 \times 88 \text{ kg} \times 9.81 = 259 \text{ N}$ [A6].
- Design drag 2,748 N [A7]; with the roller loss and 200 N of back tension the rope pull at the drum is 3,004 N [A8], equal to the rated 3 kN.

If the measured drag is higher, the skids are raised to take a thinner layer, the heap is smaller, or the rake is shortened; the capstan itself has margin (Section B).

■ B. Capstan: crank force, pull and speed (R2, R3, R12)

- Gear ratio 84 to 14, 6 to 1 [B1]. At the end of a 30 m pass the rope is on its third layer, at 110.9 mm radius [B2].
- Crank force at the design drag: $3,004 \text{ N} \times 0.1109 \text{ m} / (6 \times 0.94) / (2 \times 0.33 \text{ m}) = 89 \text{ N}$ per handle [B3]; R2 asks for 120 N or less.
- Pull from two people at 120 N: 4,030 N on the third layer [B4] and 3,685 N on a full fourth layer [B5]; R3 asks for 3 kN.
- Rope speed at 30 crank turns a minute: 0.053 m/s, about 3.1 m/min [B6]; each cranker works at 84 W [B7].
- Rope stored: 58 m in four layers of 22 turns [B8], with 38 mm of flange above the fourth layer [B9], which is more than two rope diameters. R12 asks for 45 m.

■ C. Strength at twice the rated pull (R3, R10, R11)

Table 2. Strength checks at 6 kN rope pull.

Part	Load case	Stress or force	Margin
Drum shaft, 40 mm C45	Rope load mid-span between bearings 452 mm apart, plus torque on a full drum	147 MPa combined [C1]	2.3 on yield [C1b]
Crank shaft, 25 mm C45	Tooth force on the pinion, 45 mm outside the bearing, plus torque	161 MPa [C2]	2.1 [C2b]
Pinion teeth, module 4, 35 mm face	Lewis bending, 14 teeth	139 MPa [C3]	2.5 [C3b]
Pawl and its boss	7.8 kN at the tooth [C4]; 40 mm boss welded to the cheek, 52 mm long	59 MPa [C5]	4.7 [C5b]
Brake band	247 degrees of wrap [C6]	92 N on the lever holds the rated pull [C7]	R11 asks for 150 N or less
Rope	3 kN rated pull, splices at 85 %	27 kN breaking strength	7.7 working-load factor [C8]
Chains, 8 mm grade 30	Two chains at 23 degrees down to the anchors	3.26 kN each [C9]	2.4 on working load limit [C9b]
Ground anchors, 150 mm helix	Firm clay, 50 kPa	7.95 kN holding each [C10]	2.4 on the chain force [C10b]
Rake beam, FRP 100 x 100 x 6	Heap and layer load along 3 m, pulled at both ends	28 MPa [C13]	7.1 [C13b]; bows 11 mm at the rated load [C14]

The pawl was first drawn on a plain 16 mm pin cantilevered from the cheek; at 6 kN that pin would have bent (about 580 MPa), so the constructable design carries the pawl on a 40 mm boss welded to the cheek with an M24 bolt through it (SDG-DDR-002).

Tipping: the capstan weighs 102 kg without its anchors [C11]. With the rope leaving the top of the drum at 677 mm and the chains fixed at 650 mm, the resisting moment about the front tips of the runners is 1.55 times the tipping moment at the rated pull [C12].

Safety: These margins hold only with certified rope, chain and shackles of the stated sizes, anchors that pass their on-site proof load, and parts made from the stated steels. Each is a safety stop in the build plan.

■ D. Floor, lift-off near the bund, rate and set-up (R1, R4, R5, R8)

- Floor: with the ballast the skids press on the floor at 10.7 kPa [D1], less than a person standing; the blade edge stays 10 mm above the floor [D2] and no steel touches the pan.
- Lift-off near the bund: the rope comes down from the roller top, 156 mm above the crest, to the bridle swivel 130 mm above the floor. As the rake nears the bund the rope steepens, and once its upward pull exceeds the rake's weight the blade lifts and the heap is left behind. Without ballast this happens with the blade 5.9 m from the bund toe [D3]; with two 24 kg bags of salt it moves in to 3.3 m [D4]. The windrow therefore lies about 3.3 m from the toe, and loading it out uses existing methods.
- One strip: winding a 30 m pass 9.5 min [D5], winding the empty rake back 5.7 min [D6], moving 8 min; 23.2 min in all [D7].
- Gathering rate: one full heap per strip gives about 598 kg/h [D8], or 150 kg per worker-hour for a crew of four [D9]. R5 compares this with the dantala on the same pan; that baseline is not yet measured, so R5 is not verified.
- Set-up per crew, both crews working at once: carry the capstan into place 5 min, two anchors 6 min, chains 2 min, roller 1 min, rope round the pan to the rake 5 min: 19 min [D10], just inside R8.

■ E. Mass and cost (R6, R9)

- Heaviest single part: a ballast bag at 23.8 kg, filled on site [E1]; the heaviest made parts are the drum (21.3 kg) and a side frame (20.9 kg). An assembled capstan weighs 102 kg; it is slid along the bund on its runners and is taken apart for moves between pans.
- Parts cost of the full set: USD 2,803 [E2]. Value-engineering target: USD 4,000. Estimated cost of the constructable design: USD 2,803 (USD 1,197 under the target) [E3].

■ F. Results against every requirement

Table 3. Results against SDG-REQ-001.

ID	Target	Result	Status
R1	No worker enters the pan	Crews on dry ground; rake re-aligned by the return pull	Met by design
R2	120 N or less per handle	89 N [B3]	Met
R3	3 kN from two people at 120 N; margin 2 at 6 kN	4.0 and 3.7 kN [B4], [B5]; lowest margin 2.1 [C2b]	Met
R4	Edge 5 to 25 mm up; no steel in the pan; 15 kPa or less	10 mm; 10.7 kPa [D1], [D2]	Met by design
R5	At least the dantala rate	150 kg per worker-hour [D9]; no baseline	Not verified
R6	Heaviest part 25 kg or less	23.8 kg [E1]	Met
R7	One season in brine	Material choice	Met by design
R8	20 min or less	19 min [D10]	Met, small margin
R9	Value-engineering target USD 4,000	USD 2,803 [E2]	USD 1,197 under the target
R10	Working-load factor 5 or more	7.7 [C8]	Met
R11	Pawl 6 kN; brake 150 N or less; anchors 1.5 times their share	4.7, 92 N, 2.4 [C5b], [C7], [C10b]	Met on paper; soil to confirm
R12	45 m of rope per drum	58 m [B8]	Met

■ Checks against the TRL 2 figures

The scaffold set a 3 kN pull (R3) before any drag estimate; the drag estimate here lands on it, which fixed the rake at 3.0 m and the blade at 300 mm. The scaffold had no figure for the windrow position; the lift-off calculation added the ballast bags and moved the windrow from the bund edge to about 3.3 m out (SDG-DDR-002).